D-modules
Lectures by Victor Ginzburg

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Preface

There will be some gaps in explanation, either due to the lecturer's admission or my own lack of understanding. In particular, many "proofs" are sketches of proofs. Gaps due to my own misunderstanding will be indicated by three red question marks: ???. More generally, my own questions about the material will also be in red. Things like "**Question**" will be questions posed by the lecturer. Feel free to reach out to me with explanations.

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D-modules are modules over rings of differential operators. There are three settings:

1. C^∞ case: We can consider differential operators on $\mathbb{R}^n$ of order N, which are expressions

$$u = \sum_{i_1+\cdots+i_n \le N} a_{i_1,\ldots,i_n}(x) \frac{\partial^N}{\partial x_1^{i_1} \cdots \partial x_n^{i_n}},$$

   where the functions $a_\bullet(x)$ are smooth.

2. Complex analytic setting: Similar, but replace $\mathbb{R}$ with $\mathbb{C}$ and require the functions $a_\bullet(x)$ are holomorphic.

3. Algebraic setting: Replace $\mathbb{C}$ with any field and require the functions $a_\bullet(x)$ to be polynomials.

We will work with smooth algebraic varieties over $\mathbb{C}$.

Remark. 1. If X is singular, the ring $\mathcal{D}(X)$ of differential operators might not be noetherian.

2. The theory of differential operators in positive characteristic is very different.

3. We can effectively replace $\mathbb{C}$ by any algebraically closed field of characteristic zero.

Today X is smooth and affine. $\mathcal{O} = \mathcal{O}(X)$ will denote the algebra of regular functions on X. We might not have coordinates on X, so we need to modify our definition of differential operators.

Differential operators of order 0 are just multiplication by some $f \in \mathcal{O}$, denoted m_f. Differential operators of order 1 are $\xi + m_f$, where $f \in \mathcal{O}$ and ξ is a vector field on X, i.e. an element of $T_X = \mathrm{Der}(\mathcal{O}, \mathcal{O})$, i.e. a derivation $\mathcal{O} \to \mathcal{O}$.

Definition 1.1. The algebra $\mathcal{D} = \mathcal{D}(X)$ of differential operators on X is defined as the subalgebra of $\mathrm{End}_{\mathbb{C}}(\mathcal{O})$ generated by the operators of order 0 and 1 defined above.

Note that $\mathcal{O} + T_X$ sits as a subspace in $\mathcal{D}$.

When $X = \mathbb{C}^n$, or more generally is an open subvariety of $\mathbb{C}^n$, then this definition agrees with the previous one using the induced coordinates from $\mathbb{C}^n$.

1.1 Applications of D-modules

Fix $f \in \mathcal{O}$. Let $\mathrm{Ann}(f) = \{u \in \mathcal{D} \mid u(f) = 0\}$. It is a left ideal in $\mathcal{D}$. We define $\mathcal{D}f$ to be $\mathcal{D}/\mathrm{Ann}(f)$, which is a $\mathcal{D}$-module.

We can define $\mathcal{D}f$ for some $f \notin \mathcal{O}$:

1. Let $X = \mathbb{C}$ and $f = e^x$. Certainly f is not regular, but its annihilator is generated by $\frac{d}{dx} - 1$, so we can define $\mathcal{D}e^x = \mathcal{D}/(\frac{d}{dx} - 1)$.

2. Similarly we can define $\mathrm{Ann}(f)$ and hence $\mathcal{D}f$ for $X = \mathbb{C}$ and $f \in C^\infty(\mathbb{R})$.

3. Let $f \in \mathbb{R}[x_1, \ldots, x_n]$ and let $U = \{x \in \mathbb{R}^n \mid f(x) \geq 0\}$. Fix $\varphi \in C^\infty(U)$ with compact support. Let $I_\phi(\lambda) = \int_U \varphi(x) f(x)^\lambda dx$ for $\lambda \in \mathbb{C}$. Note that the exponentiation is well-defined because $f(x) \geq 0$. The integral converges when the real part of λ is positive. Then

   Theorem 1.1. *The function $\lambda \mapsto I_\varphi(\lambda)$ extends to a meromorphic function on $\mathbb{C}$ with poles at a finite union of arithmetic progressions of the form $a, a-1, \ldots$.*

   Sketch. Say $I_\varphi(\lambda)$ converges for $\lambda > N$ (at the start, we have $N = 0$, but we proceed inductively). We compute

   $$\frac{d}{d\lambda} I_\varphi(\lambda) = \int_U \varphi(x) \log f(x) f(x)^\lambda dx,$$

   which converges, so $I_\varphi(\lambda)$ is holomorphic when $\mathrm{Re}\lambda \geq N$. Our goal is to extend the function to $\mathrm{Re}\lambda \geq N - 1$ inductively. We need:

   Lemma 1.1 (b-function)**.** *There exists $u \in \mathcal{D}(\mathbb{C}^n)[s]$ and $b \in \mathbb{C}[s]$ such that $u(f^{s+1}) = b(s) f^s$.*

   Assuming the lemma, we have

   $$\begin{aligned} I_\varphi(\lambda) = \int_U \varphi(x) f(x)^\lambda dx &= \int_U \varphi(x) \frac{1}{b(\lambda)} u(f(x)^{\lambda+1}) dx \\ &= \frac{1}{b(\lambda)} \int_U (u^*(\varphi))(x) f(x)^{\lambda+1} dx, \end{aligned}$$

   where at the end we have used essentially integration by parts, and $u^* \in \mathcal{D}(\mathbb{C}^n)[s]$ is the formal adjoint of u. For $\mathrm{Re}\lambda > N - 1$ we can extend $I_\varphi(\lambda)$ using the above computation.

Example 1.1 (Examples of b-functions)**.**

1. Let $f(x) = \sum x_i^2$ on $\mathbb{C}^n$. Then we can take $u = \sum \frac{\partial^2}{\partial x_i^2}$ and $b(s) = 4(s+1)(s+\frac{n}{2})$.

2. Let $f(x) = \det(x)$ on $\mathbb{C}^{n^2} = M_n(\mathbb{C})$. Then for $u = \det(\frac{\partial}{\partial x_{ij}})$ one can compute $u(f^s)$ using Capelli identities and find that $b(s) = \prod_{1 \leq k \leq n} (s+k)$.

3. If $f(x_1, \ldots, x_n) = \prod_{1 \le i < j \le n}$, then the b-function is unknown. Can also generalize to arbitrary root systems.

4. If one takes the square of the above function, then the b-function is known, but is very complicated.

1.2 Nearby cycles

Let $f \in \mathbb{C}[x_1, \ldots, x_n]$. Let $\mathrm{Sing}(f) = \{x \in \mathbb{C}^n \mid df(x) = 0\}$. Fix $z_0 \in \mathrm{Sing}(f)$. Take $0 < \delta \ll \varepsilon \ll 1$. We define the *link* of f at z_0 to be

$$L = L(f, z_0, \varepsilon, \delta) = \{z \in \mathbb{C}^n \mid \|z - z_0\| \le \varepsilon, \|f(z) - f(z_0)\| = \delta\}.$$

Topologically, L is independent of δ, ε. there's a nice picture but seems hard to TiKz. Let S^1_δ denote the circle of w such that $\|w - f(z_0)\| = \delta$. The family $H_p(L_w)$ forms a locally constant sheaf on S^1_δ. For fixed w, there is a *monodromy operator* $M_{f,p} : H_p(L_w) \to H_p(L_w)$.

Theorem 1.2. *If $b_f(\lambda) = 0$, then $e^{2\pi i \lambda}$ is an eigenvalue of $M_{f,p}$ for some p. Moreover if $\mathrm{Sing}(f) = \{z_0\}$, then all eigenvalues of all $M_{f,p}$ arise in this way.*

1.3 Microlocal geometry

$\mathcal{D}(X)$ is a noncommutative deformation (aka quantization) of $\mathcal{O}(T^*X)$.

For $f \in \mathcal{O}(X)$, we have $\mathrm{Sing}(f) \subset X$. We also have the $\mathcal{D}$-module $\mathcal{D}f$. To a $\mathcal{D}$-module we can define its singular support, which sits in T^*X. The following square commutes:

$$\begin{tikzcd} SS(\mathcal{D}f) \arrow[r, hook] \arrow[d] & T^*X \arrow[d] \\ \mathrm{Sing}(f) \arrow[r, hook] & X \end{tikzcd}$$

The singular support of $\mathcal{D}f$ can contain more information than just $\mathrm{Sing}(f)$.

1.4 de Rham functor

There is a functor from the (derived?) category of $\mathcal{D}(X)$-modules to the (derived?) category of complexes of sheaves of $\mathbb{C}$-vector spaces on X. It restricts to a functor between the (derived?) category of holonomic modules to the category of perverse sheaves. This further restricts to an equivalence of categories between holonomic modules with regular singularities (or complexes with regular holonomic cohomology?) to the category of perverse sheaves.

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2.1 Poincaré-Birkhoff-Witt (PBW) for differential operators

We continue to assume X is smooth affine, and we write $\mathcal{O}, \mathcal{D}$ for $\mathcal{O}(X), \mathcal{D}(X)$. We let T_X denote the space of vector fields.

For $\xi \in T_X$ and $f, g \in \mathcal{O}$, we have the Liebniz formula $\xi(fg) = f\xi(g) + \xi(f)g$. In particular, in terms of operators, $\xi m_f - m_f \xi = m_{\xi(f)}$. Also, for $\eta \in T_X$, we have $\xi\eta - \eta\xi = [\xi, \eta]$, where $[-,-]$ is the (a priori abstract) Lie bracket of vector fields. It follows that $\mathcal{D}_1 = \mathcal{O} + T_X$ is a Lie algebra. $\mathcal{O}$ is an abelian Lie ideal and T_X is a Lie subalgebra. We have a short exact sequence $0 \to \mathcal{O} \to \mathcal{D}_1 \to T_X \to 0$. The formula $\xi m_f - m_f \xi = m_{\xi(f)}$ also shows that $\mathcal{D}_1$ is an $\mathcal{O}$-bimodule, and that the left and right actions agree on $\mathcal{D}_1/\mathcal{O}$.

Definition 2.1. The **order filtration** on $\mathcal{D}$ is given by $\mathcal{D}_0 = \mathcal{O}$, $\mathcal{D}_1 = \mathcal{O} + T_X$, and $\mathcal{D}_n$ is the image of the composition map $\mathcal{D}_1 \otimes_{\mathcal{O}} \cdots \otimes_{\mathcal{O}} \mathcal{D}_1 \to \mathcal{D}$.

We have

- $\bigcup_n \mathcal{D}_n = \mathcal{D}$.
- $\mathcal{D}_n$ is an $\mathcal{O}$-bimodule.
- $\mathcal{D}_m \mathcal{D}_n \subset \mathcal{D}_{m+n}$.

Let $\mathrm{gr}\mathcal{D} = \bigoplus(\mathcal{D}_n/\mathcal{D}_{n-1})$. Let $f \in \mathcal{O}$ and $\xi, \eta \in T_X$. Then $f\xi \equiv \xi f \mod \mathcal{O}$, so $\mathcal{O} = \mathcal{D}_0$ is a central subalgebra of $\mathrm{gr}\mathcal{D}$. Similarly, $\xi\eta \equiv \eta\xi \mod \mathcal{D}_1$, so $T_X = \mathcal{D}_1/\mathcal{D}_0$ is contained in $Z(\mathrm{gr}\mathcal{D})$. We obtain a surjective graded algebra map $\mu : \mathrm{Sym}_{\mathcal{O}} T_X \to \mathrm{gr}\mathcal{D}$.

Theorem 2.1 (PBW for $\mathcal{D}$)**.** *μ is an isomorphism.*

Proof. It suffices to show that μ is injective.

The cotangent bundle T^*X is a vector bundle, so it admits an (attracting) algebraic $\mathbb{C}^\times$-action given by fiberwise scaling. This gives a grading $\mathcal{O}(T^*X) = \bigoplus_{n \geq 0} \mathcal{O}_n(T^*X)$, where $\mathcal{O}_n(T^*X)$ consists of functions that are fiberwise homogeneous of degree n. Now, there is a canonical isomorphism $\nu : \mathrm{Sym}^n_{\mathcal{O}}(T_X) \xrightarrow{\sim} \mathcal{O}_n(T^*X)$, where a product of vectors $\xi_1 \cdots \xi_n$ is sent to the function taking $(x, \alpha) \in T^*X$ to $\prod \langle (\xi_i)_x, \alpha \rangle$.

Now, there is a so-called principal symbol map that we will define in a moment, $\sigma_n : \mathcal{D}_n \to \mathcal{O}_n(T^*X)$, that sends $\mathcal{D}_{n-1}$ to 0, so it descends to a map $\overline{\sigma}_n : \mathcal{D}_n/\mathcal{D}_{n-1} \to \mathcal{O}_n(T^*X)$. We will check that the composite

$$\mathrm{Sym}^n_{\mathcal{O}} T_X \xrightarrow{\mu} \mathcal{D}_n/\mathcal{D}_{n-1} \xrightarrow{\overline{\sigma}_n} \mathcal{O}_n(T^*X) \xrightarrow{\nu^{-1}} \mathrm{Sym}^n_{\mathcal{O}} T_X$$

is the identity, which implies μ is injective.

Now we construct σ_n. Note that $\mathcal{D}$ acts on $\mathcal{O}[[t]]$ by $u(\sum f_i t^i) = \sum u(f_i)t^i$. For fixed $f \in \mathcal{O}$, we have $e^{tf} = \sum \frac{1}{i!} f^i t^i \in \mathcal{O}[[t]]$.

Claim 2.1. *1. For $u \in \mathcal{D}_n$, we have*

$$u(e^{tf}) = (t^n \sigma_n(u, f) + \text{ lower degree terms})\, e^{tf},$$

where $\sigma_n(u, f), A \in \mathcal{O}$.

2. If $u \in \mathcal{D}_{n-1}$, then $\sigma_n(u, f) = 0$.

3. We have

$$\sigma_n(\xi_1 \cdots \xi_n, f)(x) = \prod \langle (\xi_i)_x, d_x f \rangle = \nu(\xi_1 \cdots \xi_n)(d_x f).$$

Proof of Claim. First we have

$$\xi e^{tf} = t\xi(f)e^{tf} = t\langle \xi, df \rangle e^{tf}.$$

Then we have

$$\xi_1\xi_2 e^{tf} = t^2 \xi_1(f)\xi_2(f)e^{tf} + t\xi_1\xi_2(f)e^{tf} = \left(t^2 \langle \xi_1, df \rangle \langle \xi_2, df \rangle + t\xi_1\xi_2(f)\right) e^{tf}.$$

And from here the pattern is clear.

This claim both gives the construction of σ_n and shows that the previous composition is the identity, so we are done. I want to check this in more detail I think

Remark. PBW and its proof work in the C^∞ or complex analytic settings.

2.2 An application to the PBW theorem for Lie algebras

Let $\mathfrak{g}$ be a Lie algebra, and let $\mathcal{U}\mathfrak{g}$ be its universal enveloping algebra. Lecturer: "since this is not a course on Lie algebras, I'm not supposed to explain what it is." $\mathcal{U}\mathfrak{g}$ has a PBW filtration defined in a very similar way to the order filtration. In particular $\mathcal{U}_0\mathfrak{g} = \mathbb{C}$ and $\mathcal{U}_1\mathfrak{g} = \mathbb{C} + \mathfrak{g}$. Furthermore, the relations of $\mathcal{U}\mathfrak{g}$ imply that we get a surjective graded algebra map $\mu : \mathrm{Sym}_{\mathbb{C}}\mathfrak{g} \to \mathrm{gr}\mathcal{U}\mathfrak{g}$. The PBW theorem (for Lie algebras) asserts that μ is an isomorphism. It is not easy, but if we assume that $\mathfrak{g}$ is the Lie algebra of a Lie group or affine algebraic group $\mathfrak{g}$, then it is easy.

Note. It is in fact true that every $\mathfrak{g}$ comes from a G, but the proof is no easier than PBW, and in some cases one uses PBW in the proof.

We can identify $\mathfrak{g}$ with $(T_G)^G$, i.e. left-invariant vector fields on G, as well as with T_eG. (At least one of these is the definition of Lie$G$, depending on your conventions.) We get a map $\mathfrak{g} \to \mathcal{D}(G)^G$ by sending a vector field to its corresponding differential operator, which gives a map $\tau : \mathcal{U}\mathfrak{g} \to \mathcal{D}(G)^G$.

Theorem 2.2. *1. The map τ is an algebra isomorphism.*

2. PBW for $\mathcal{U}\mathfrak{g}$ is true.

Proof. By construction, $\tau(\mathcal{U}_n\mathfrak{g}) \subset \mathcal{D}_n(G)^G$, so there is an induced graded map $\mathrm{gr}(\tau) : \mathrm{gr}\mathcal{U}\mathfrak{g} \to \mathrm{gr}\mathcal{D}(G)^G$. We then have the following commutative diagram:

$$\begin{tikzcd}
\mathrm{Sym}^n_{\mathbb{C}}\mathfrak{g} \arrow[r, "\sim"] \arrow[d, two heads, "\mu_n"'] & \mathrm{Sym}^n_{\mathcal{O}}((T_G)^G) \arrow[r, "\sim"] & (\mathrm{Sym}^n_{\mathcal{O}}T_G)^G \arrow[d, "\sim\text{by PBW for } \mathcal{D}"] \\
\mathcal{U}_n\mathfrak{g}/\mathcal{U}_{n-1}\mathfrak{g} \arrow[r, "\mathrm{gr}_n(\tau)"'] & \mathcal{D}_n(G)^G/\mathcal{D}_{n-1}(G)^G \arrow[r, hook] & (\mathcal{D}_n(G)/\mathcal{D}_{n-1}(G))^G
\end{tikzcd}$$

The right map on the bottom is injective because $\mathcal{D}_n(G)^G \cap \mathcal{D}_{n-1}(G) = \mathcal{D}_{n-1}(G)^G$. It follows that all arrows are isomorphisms, so we get that $\mathrm{Sym}_{\mathbb{C}}\mathfrak{g} \to \mathrm{gr}\mathcal{U}\mathfrak{g}$ and $\mathrm{gr}\mathcal{U}\mathfrak{g} \to \mathrm{gr}\mathcal{D}(G)^G$ are both isomorphisms. The former isomorphism is PBW, and the latter isomorphism implies τ is an isomorphism, so we are done.

We now discuss another formulation of $\mathcal{D}(X)$. By the lecturer's admission, this may be wrong as it was done on the spot. We have $\mathcal{D}(X) = \mathcal{U}_{\mathbb{C}}(\mathcal{O}+T_X)/(f\otimes\xi = f\xi; 1_{\mathcal{D}} = 1_{\mathcal{O}} = 1_{\mathcal{U}})$. The proof uses PBW to show that the associated graded algebras agree.